Curriculum Vitae

Daniel Rogozin
Research Scientist in Mathematical Logic, Programming Languages, Formal Methods and AI

I'm an interdisciplinary research scientist and functional software engineer working at the intersection of programming-language theory, formal methods and AI. My work combines category theory, type theory and mathematical logic with practical development in Haskell, Lean 4 and compiler-related systems. I have contributed to program-synthesis tooling, cryptographic formalisation, zero-knowledge systems, blockchain infrastructure and GHC, and have published and presented research in logic and theoretical computer science.

1 Technical Skills

- Research in mathematics: mathematical logic and category theory, type theory, proof theory, constructive mathematics,
- Proof assistants and formal verification (Agda, Rocq, Lean),
- Functional programming (Haskell, OCaml, Rust),
- Applications: compiler development, program synthesis, knowledge representation, zero-knowledge proofs, cryptography, blockchain systems.

2 Employment

- (March 26 — June 26) Corca Research, Mathematician, project-based engagement.

  The development of the prototype autoformalising mathematical content of a text written in a natural language.

- (Sep' 23—Feb' 26) Noeon Research, Research Scientist.

  I contributed to a number of internal projects related to type theory and category theory with applications to interpretable artificial intelligence and knowledge representation:

  - Formalising the category theory-based reasoning framework for program synthesis with linear type theory.
  - Developing a Haskell type checker for the DreamCoder-based program synthesis utility.
  - Reviewing current literature in programming language theory and symbolic methods,
  - Preparing educational lectures in mathematical logic, analytic philosophy of language, denotational semantics, universal algebra, type theory, programming language theory, etc.

- (Jan' 23—June 23) University College London, Department of Computer Science, teaching assistant.
- (May' 22—April' 23) Yatima Inc/Lurk Lab, research formalisation engineer. The projects include:
  - The Lean 4 formalisation of finite fields, elliptic curve cryptography and zero-knowledge proof generation frameworks,
  - Contributing to Lurk (both Rust and Lean implementations), a LISP-based programming language with compilation into circuits over finite fields,
  - Preparing the experience report for International Conference on Functional Programming 2023.
- (Feb' 18—Jun' 22) Serokell OÜ, Tallinn, Estonia, Senior Haskell software engineer. The projects include:
  - (2018) Cardano, a cryptocurrency backend in Haskell.

- (2019) Machine Learning in Haskell, an R&D project on type-safe matrix operations via singletons. The work on that project also included the preparation of blogposts and the conference talk.
- (2019) Formality, an R&D project, a dependently typed language with affine types.
- (2020–2022) Glasgow Haskell Compiler contribution: the implementation of the Char Kind and partial implementation of @-binders in function declarations.

- (Jan' 21—Mar' 22) Institute for Information Transmission Problems of Russian Academy of Sciences, Laboratory of Algebra and Number Theory, Trainee researcher.

3 Education

- PhD programme at the Institute for Information Transmission Problems (2020–2022) under supervision by Ilya Shapirovsky [1].
- Master's degree at the Lomonosov Moscow State University, The Faculty of Philosophy (2016–2018).
- Bachelor's degree at the State Academic University for the Humanities, The Faculty of Philosophy (2012–2016).

4 Theses

- 2018, *Modal type theory based on intuitionistic epistemic logic*, Master's thesis, Lomonosov Moscow State University.
- 2016, *Type theory in the foundations of mathematics*, Bachelor's thesis, State Academic University for the Humanities.

5 Publications and Preprints

- Daniel Rogozin. "Intuitionistic Linear Logic with Subexponentials: Type Theory, Categorical Models and Realisability." arXiv preprint arXiv:2507.12360 (2025).
- Daniel Rogozin. "Semantic Analysis of Subexponential Modalities in Distributive Non-commutative Linear Logic", *Proceedings Modalities in substructural logics: Applications at the interfaces of logic, language and computation (AMSLO 2023)*, *Electronic Proceedings in Theoretical Computer Science*, 381, pp. 60–70.
- Daniel Rogozin and Ilya Shapirovsky. "On decidable extensions of Propositional Dynamic Logic", *Journal of Applied Logics – IfCoLog Journal*, 10(4): 629–642, 2023.
- Amin, Nada, John Burnham, François Garillot, Rosario Gennaro, Daniel Rogozin, Chhi'med Kunzang and Cameron Wong. "Lurk: Lambda, the ultimate recursive knowledge." *Proceedings of the ACM on Programming Languages*, 7(ICFP), 259-274.
- Daniel Rogozin. "Some results on relation algebra reducts: Residuated and semilattice-ordered semigroups." *Journal of Logic and Computation* 32, (2022): 266–280.
- Daniel Rogozin. "Reducts of relation algebras: The aspects of axiomatisability and finite representability." In *International Symposium on Logical Foundations of Computer Science*, pp. 266–280. Springer, Cham, 2022.
- Daniel Rogozin. "Categorical and algebraic aspects of the intuitionistic modal logic IEL$^-$ and its predicate extensions." *Journal of Logic and Computation* 31, no. 1 (2021): 347–374.
- Daniel Rogozin. "Modal Type Theory Based on the Intuitionistic Modal Logic IEL$^-$." In *International Symposium on Logical Foundations of Computer Science*, pp. 236–248. Springer, Cham, 2020.

[1] Unfinished because of the political circumstances in Russia

6 Conference Talks

- Winter 2026, workshop "Formalisation and deformalisation of mathematical reasoning", Paris, France, *Operational Knowledge Representation with Monoidal Categories and Linear Type Theory.*
- September 2025, conference "British Logic Colloquium 2025", Manchester, UK, *Term Assignment and Categorical Models for Intuitionistic Linear Logic with Subexponentials.*
- July 2023, 34th European Summer School in Logic, Language and Information, workshop "Modalities in substructural logics: applications at the interfaces of logic, language and computation", Ljubljana, Slovenia, *Semantic Analysis of Subexponential Modalities in Distributive Non-commutative Linear Logic.*
- August 2022, conference "Advances in Modal Logic 2022", Rennes, France, *On decidable extensions of Propositional Dynamic Logic* (joint work with Ilya Shapirovsky).
- January 2022, conference "Logical Foundations of Computer Science 2022", Deerfield Beach, Florida, USA, *Reducts of Relation Algebras: The Aspects of Axiomatisability and Finite Representability.*
- June 2021, conference "BLAST: Boolean Algebras, Lattices, Algebraic and Quantum logic, Universal Algebra, Set Theory Set-theoretic and Point-free Topology", New Mexico State University, New Mexico, USA, *Reducts of representable relation algebras: finite representability and axiomatisability.*
- August 2020, conference "Advances in Modal Logic 2020", Helsinki, Finland, *Canonical Extensions for the Distributive Full Lambek Calculus with Modal Operators.*
- January 2020, conference "Logical Foundations of Computer Science 2020", Deerfield Beach, Florida, USA, *Modal Type Theory Based on the Intuitionistic Modal Logic* $\mathbf{IEL}^-$.
- June 2019, conference "Topology, Algebra, and Categories in Logic", Nice, France, *Quantale semantics for Lambek calculus with subexponentials.*

7 Seminar Talks

- Winter 2023, PPLV group seminar, UCL, London, *Intuitionistic epistemic logic categorically and algebraically.*
- Winter 2023, PPLV group seminar, UCL, London, *Decidable extensions of propositional dynamic logic.*
- Spring 2022, New Mexico State University, New Mexico, *Categories and lambda calculus for programming language theory.*
- Autumn 2021, seminar "Computational Logic", City University of New York, The Graduate Centre, *Decidability results for relation algebras and their reducts.*
- Autumn 2020, seminar "Computational Logic", City University of New York, The Graduate Centre, *Categorical and Algebraic Aspects of the Intuitionistic Modal Logic* IEL^- *and its predicate extensions.*

8 Teaching

- Winter and Spring 2023, University College London, The Department of Computer Science, London, UK, *Introduction to Logic and Algorithms*, teaching assistant.
- (Spring 2019 – Autumn 2021), Higher School of Economics, The Faculty of Computer Science, Moscow, *Functional programming* (with Stepan Kuznetsov), lecturer and teaching assistant.
- (Spring 2019 – Autumn 2021), ITMO University, The Department of CS, St. Petersburg, *Functional Programming*, lecturer and teaching assistant.
- Autumn 2021, Computer Science Club, St. Petersburg, *Semantic aspects of intuitionistic logic*, lecturer.
- Autumn 2019, Computer Science Club, St. Petersburg, *Introduction to modal logic*, lecturer.
- Summer 2019, Computer Science Summer School, Narva, Estonia, lecture *Introduction to computability and logic.*

9 Blogposts

- A Walk Through L: Operationalising Knowing How with Category Theory and Linear Logic,
- How Dependent Haskell Can Improve Industry Projects (written with Vladislav Zavialov),
- Formal verification: History and Methods,
- Incomplete and Utter Introduction to Modal Logic, Pt. 1
- Incomplete and Utter Introduction to Modal Logic, Pt. 2,
- Dimensions and Haskell: Introduction,
- Dimensions and Haskell: Singletons in Action,
- Constructive and Non-Constructive Proofs in Agda (Part 1): Logical Background,
- Constructive and Non-Constructive Proofs in Agda (Part 2): Agda in a Nutshell,
- Constructive and Non-Constructive Proofs in Agda (Part 3): Playing with Negation.

10 Awards

- 2022, Best student paper award, for the paper *Reducts of relation algebras: The aspects of axiomatisability and finite representability* at Logical Foundations of CS 2022, Miami.
- 2020-2021, Category "Best Course for New Knowledge and Skills" for the Functional programming course at Higher School of Economics, Moscow.

11 Languages

Russian (native speaker), English (fluent, close to native), French (actively learning, currently A2), Yiddish (learning for cultural curiosity).

12 Contact Info and Links

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- Google Scholar: `https://scholar.google.com/citations?user=t3Vf1oAAAAAJ&hl=en`